

INTERNET TECHNOLOGY & WEB PAGE DESIGN

Unit-III: HTML (Hyper Text Markup Language)

Comprehensive Study Notes

HPU BCA Semester IV

1. Introduction to HTML

1.1 What is HTML?

HTML (Hyper Text Markup Language) is the standard markup language for creating web pages and web applications.

Definition: HTML is the standard markup language used to create web pages. It describes the structure of a web page using markup tags.

1.2 Basic HTML Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

Output: Displays "Welcome to HTML" as a heading and "This is my first web page." as a paragraph.

2. Heading Tags, Paragraph & Text Formatting

2.1 Heading Tags (h1 to h6)

HTML provides six levels of headings, from `<h1>` (largest) to `<h6>` (smallest).

```
<!DOCTYPE html>
<html>
<body>
  <h1>Heading 1 (Largest)</h1>
  <h2>Heading 2</h2>
  <h3>Heading 3</h3>
  <h4>Heading 4</h4>
  <h5>Heading 5</h5>
  <h6>Heading 6 (Smallest)</h6>
</body>
</html>
```

2.2 Paragraph and Text Formatting

```
<!DOCTYPE html>
<html>
<body>
  <p>This is a normal paragraph.</p>
  <b>This is bold text.</b><br>
  <i>This is italic text.</i><br>
  <u>This is underlined text.</u><br>
  <strong>This is strong text.</strong><br>
  <em>This is emphasized text.</em><br>
  <mark>This is highlighted text.</mark><br>
  <small>This is smaller text.</small><br>
  <del>This is deleted text.</del><br>
  <ins>This is inserted text.</ins><br>
  H<sub>2</sub>O (Subscript)<br>
  E = mc<sup>2</sup> (Superscript)
```

```
</body>  
</html>
```

2.3 Text Formatting Tags Summary

Tag	Description	Example
	Bold text	Bold
<i>	Italic text	<i>Italic</i>
<u>	Underlined text	<u>Underlined</u>
	Important text	Important
	Emphasized text	Emphasized
<mark>	Highlighted text	<mark>Highlighted</mark>
<small>	Smaller text	Small
	Deleted text	Deleted
<ins>	Inserted text	<ins>Inserted</ins>
<sub>	Subscript	H₂O
<sup>	Superscript	X²

2.4 HTML Comments

Comments are not displayed in the browser:

```
<!-- This is a comment -->  
<p>This text is visible.</p>
```

3. Lists in HTML

3.1 Ordered Lists

```
<!DOCTYPE html>
<html>
<body>
  <h3>Fruits (Ordered List)</h3>
  <ol>
    <li>Apple</li>
    <li>Banana</li>
    <li>Cherry</li>
    <li>Date</li>
    <li>Elderberry</li>
  </ol>
  <h3>Fruits (Roman Numerals)</h3>
  <ol type="I">
    <li>Apple</li>
    <li>Banana</li>
    <li>Cherry</li>
  </ol>
  <h3>Fruits (Alphabets)</h3>
  <ol type="A">
    <li>Apple</li>
    <li>Banana</li>
    <li>Cherry</li>
  </ol>
</body>
</html>
```

3.2 Unordered Lists

```
<!DOCTYPE html>
<html>
<body>
  <h3>Vegetables (Unordered List)</h3>
  <ul>
    <li>Carrot</li>
```

```
<li>Broccoli</li>
<li>Spinach</li>
<li>Tomato</li>
<li>Potato</li>
</ul>
<h3>Vegetables (Square Bullets)</h3>
<ul type="square">
  <li>Carrot</li>
  <li>Broccoli</li>
  <li>Spinach</li>
</ul>
<h3>Vegetables (Circle Bullets)</h3>
<ul type="circle">
  <li>Carrot</li>
  <li>Broccoli</li>
  <li>Spinach</li>
</ul>
</body>
</html>
```

3.3 Nested Lists

```
<!DOCTYPE html>
<html>
<body>
  <h3>Nested List</h3>
  <ul>
    <li>Fruits
      <ul>
        <li>Apple</li>
        <li>Banana</li>
      </ul>
    </li>
    <li>Vegetables
      <ul>
        <li>Carrot</li>
        <li>Broccoli</li>
      </ul>
    </li>
    <li>Grains</li>
  </ul>
```

```
</body>  
</html>
```

3.4 Description Lists

```
<!DOCTYPE html>  
<html>  
<body>  
  <h3>Programming Languages</h3>  
  <dl>  
    <dt>HTML</dt>  
    <dd>Hyper Text Markup Language</dd>  
    <dt>CSS</dt>  
    <dd>Cascading Style Sheets</dd>  
    <dt>JavaScript</dt>  
    <dd>Client-side scripting language</dd>  
    <dt>Python</dt>  
    <dd>General-purpose programming language</dd>  
  </dl>  
</body>  
</html>
```

4. Tables in HTML

4.1 Basic Table

```
<!DOCTYPE html>
<html>
<body>
  <h3>Student Details</h3>
  <table border="1">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Age</th>
      <th>Course</th>
    </tr>
    <tr>
      <td>101</td>
      <td>Rahul</td>
      <td>20</td>
      <td>BCA</td>
    </tr>
    <tr>
      <td>102</td>
      <td>Priya</td>
      <td>21</td>
      <td>BCA</td>
    </tr>
    <tr>
      <td>103</td>
      <td>Amit</td>
      <td>19</td>
      <td>BCA</td>
    </tr>
  </table>
</body>
</html>
```

4.2 Table with Rowspan and Colspan

```
<!DOCTYPE html>
<html>
<body>
  <h3>Time Table</h3>
  <table border="1">
    <tr>
      <th>Day</th>
      <th>10-11</th>
      <th>11-12</th>
      <th>12-1</th>
    </tr>
    <tr>
      <td>Monday</td>
      <td>Math</td>
      <td>Science</td>
      <td>Break</td>
    </tr>
    <tr>
      <td>Tuesday</td>
      <td>English</td>
      <td>Computer</td>
      <td>Break</td>
    </tr>
    <tr>
      <td>Wednesday</td>
      <td colspan="2">Lab Session</td>
      <td>Library</td>
    </tr>
  </table>
</body>
</html>
```

4.3 Table Tags Summary

Tag	Description
<table>	Defines a table
<tr>	Table row
<th>	Table header

<td>	Table data
<thead>	Table header group
<tbody>	Table body group
<tfoot>	Table footer group
<caption>	Table caption

5. Images in HTML

5.1 Inserting Images

The `<img>` tag is used to embed images in a webpage.

```
<!DOCTYPE html>
<html>
<body>
  <h3>Image Example</h3>
  
  <figure>
    
    <figcaption>This is a sample image with caption</figcaption>
  </figure>
</body>
</html>
```

5.2 Image Attributes

Attribute	Description	Example
src	Source file path	src="image.jpg"
alt	Alternate text	alt="Description"
width	Width in pixels	width="300"
height	Height in pixels	height="200"

6. Links in HTML

6.1 Creating Hyperlinks

The `<a>` tag defines a hyperlink. The `href` attribute specifies the URL.

```
<!DOCTYPE html>
<html>
<body>
  <h3>Hyperlinks</h3>
  <a href="https://www.example.com">Visit Example</a><br>
  <a href="mailto:info@example.com">Send Email</a><br>
  <a href="#section2">Go to Section 2 (Bookmark)</a><br>
  <a href="page.html" target="_blank">Open in New Tab</a>
  <h2 id="section2">Section 2</h2>
  <p>Content of section 2.</p>
</body>
</html>
```

6.2 Images as Links

```
<!DOCTYPE html>
<html>
<body>
  <a href="https://www.example.com">
    
  </a>
</body>
</html>
```

6.3 Link Attributes

Attribute	Description
href	Specifies the URL

target	Specifies where to open (_blank , _self , _parent , _top)
mailto:	Creates an email link
#id	Creates an anchor/bookmark link

7. Image Maps

7.1 Creating Image Maps

```
<!DOCTYPE html>
<html>
<body>
  <h3>Image Map Example</h3>
  
  <map name="workmap">
    <area shape="rect" coords="34,44,270,350" href="computer.html">
    <area shape="circle" coords="337,300,44" href="phone.html">
  </map>
</body>
</html>
```

8. Forms in HTML

8.1 Basic Form

```
<!DOCTYPE html>
<html>
<body>
  <h3>Registration Form</h3>
  <form action="submit.php" method="post">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name"><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email"><br>

    <label for="pwd">Password:</label>
    <input type="password" id="pwd" name="pwd"><br>

    <label>Gender:</label>
    <input type="radio" name="gender" value="male"> Male
    <input type="radio" name="gender" value="female"> Female<br>

    <label>Hobbies:</label>
    <input type="checkbox" name="hobby1" value="reading"> Reading
    <input type="checkbox" name="hobby2" value="music"> Music<br>

    <label for="country">Country:</label>
    <select id="country" name="country">
      <option value="india">India</option>
      <option value="usa">USA</option>
      <option value="uk">UK</option>
    </select><br>

    <label for="msg">Message:</label>
    <textarea id="msg" name="msg" rows="4" cols="30"></textarea><br>

    <fieldset>
      <legend>Contact Details</legend>
      <label for="phone">Phone:</label>
      <input type="number" id="phone" name="phone">
```

```
</fieldset><br>

<input type="submit" value="Submit">
<input type="reset" value="Reset">
<input type="button" value="Click Me">
</form>
</body>
</html>
```

8.2 Form Input Types

Type	Description	Example
text	Single line text	<code><input type="text"></code>
password	Password field	<code><input type="password"></code>
email	Email address	<code><input type="email"></code>
number	Number input	<code><input type="number"></code>
date	Date picker	<code><input type="date"></code>
file	File upload	<code><input type="file"></code>
checkbox	Checkbox	<code><input type="checkbox"></code>
radio	Radio button	<code><input type="radio"></code>
submit	Submit button	<code><input type="submit"></code>
reset	Reset button	<code><input type="reset"></code>
button	Generic button	<code><input type="button"></code>

8.3 Form Elements

Element	Description
<code><form></code>	Defines an HTML form

<code><input></code>	Input field
<code><select></code>	Dropdown list
<code><textarea></code>	Multi-line text input
<code><label></code>	Label for an input
<code><fieldset></code>	Groups related elements
<code><legend></code>	Caption for fieldset

9. Frames in HTML

9.1 Frameset Layout

Frames divide the browser window into multiple sections.

```
<!DOCTYPE html>
<html>
<head>
  <title>Frameset Example</title>
</head>
<frameset rows="100,*">
  <frame src="header.html" name="header">
    <frameset cols="20%,80%">
      <frame src="menu.html" name="menu">
      <frame src="content.html" name="content">
    </frameset>
  </frameset>
</frameset>
</html>
```

9.2 Iframe

The `<iframe>` tag embeds another page within the current page.

```
<!DOCTYPE html>
<html>
<body>
  <h3>Iframe Example</h3>
  <iframe src="https://www.example.com" width="600" height="400"></iframe>
</body>
</html>
```

10. Multimedia in HTML

10.1 Audio

```
<!DOCTYPE html>
<html>
<body>
  <h3>Audio Example</h3>
  <audio controls>
    <source src="audio.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>
</body>
</html>
```

10.2 Video

```
<!DOCTYPE html>
<html>
<body>
  <h3>Video Example</h3>
  <video width="640" height="360" controls>
    <source src="video.mp4" type="video/mp4">
    Your browser does not support the video element.
  </video>
</body>
</html>
```

11. Program Writing Questions

Q1. Write HTML code to create a simple webpage with heading, paragraph, and list.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Page</title>
</head>
<body>
  <h1>Welcome to My Website</h1>
  <p>This is a paragraph about web development.</p>
  <h3>Topics Covered:</h3>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
  </ul>
</body>
</html>
```

Q2. Write HTML code to create a table with student details.

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Table</title>
</head>
<body>
  <h3>Student Details</h3>
  <table border="1">
    <tr>
      <th>Roll No</th>
      <th>Name</th>
      <th>Marks</th>
    </tr>
    <tr>
      <td>101</td>
```

```
        <td>Rahul</td>
        <td>85</td>
    </tr>
    <tr>
        <td>102</td>
        <td>Priya</td>
        <td>90</td>
    </tr>
</table>
</body>
</html>
```

Q3. Write HTML code to create a registration form.

```
<!DOCTYPE html>
<html>
<head>
    <title>Registration Form</title>
</head>
<body>
    <h3>Registration Form</h3>
    <form action="submit.php" method="post">
        Name: <input type="text" name="name"><br>
        Email: <input type="email" name="email"><br>
        Password: <input type="password" name="pwd"><br>
        Gender: <input type="radio" name="gender" value="male"> Male
                <input type="radio" name="gender" value="female"> Female<br>
        <input type="submit" value="Register">
    </form>
</body>
</html>
```

12. Questions

Short Answer Questions (2-5 Marks)

1. What is HTML?
2. What are the basic tags in HTML?
3. What is the difference between ordered and unordered lists?
4. What are the attributes of the table tag?
5. What is the difference between `<th>` and `<td>` ?
6. What are image attributes?
7. What is an image map?
8. What are the form elements?
9. What are the different form input types?
10. What are frames in HTML?

Objective Questions (1 Mark)

1. HTML stands for:
(a) Hyper Text Markup Language (b) High Text Markup Language
(c) Hyper Transfer Markup Language (d) None
2. The largest heading is:
(a) `<h1>` (b) `<h6>` (c) `<h3>` (d) `<h4>`
3. The tag for bold text is:
(a) `<b>` (b) `<strong>` (c) Both (d) None
4. The table row tag is:
(a) `<tr>` (b) `<td>` (c) `<th>` (d) `<table>`
5. The form action attribute specifies:
(a) Where to send data (b) Form name (c) Form method (d) None

Quick Revision

HTML Document Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  Content goes here
</body>
</html>
```

Common HTML Tags

Category	Tags
Text	<h1> to <h6>, <p>, , <i>, <u>, , , <mark>, <small>, , <ins>, <sub>, <sup>
Lists	, , <dl>, , <dt>, <dd>
Tables	<table>, <tr>, <th>, <td>, <caption>
Links	Link
Images	
Forms	<form>, <input>, <select>, <textarea>, <label>
Frames	<frameset>, <frame>, <iframe>
Multimedia	<audio>, <video>

End of Unit-III Notes